

Dr. Aisha Williams

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PROFESSIONAL SUMMARY

Senior Data Scientist with PhD in Machine Learning and 6+ years of industry experience. Expert in deep learning, NLP, large-scale ML engineering, and production model deployment. Published 5 peer-reviewed papers at NeurIPS and ICML. Kaggle Master with 3 competition medals. Led data-science initiatives delivering \$15 M+ in measurable business impact.

WORK EXPERIENCE

AI Labs Corp | Senior Data Scientist | 2021 – Present

- Led NLP recommendation engine using fine-tuned BERT — improved click-through rate by 35% (15 M users)
- Architected production ML pipeline processing 10 M+ predictions/day (MLflow + Kubernetes + Seldon)
- Trained CNN/ResNet computer-vision models for manufacturing defect detection — 97.3% precision, 96.8% recall
- Built LSTM and Transformer time-series forecasting models reducing inventory waste by 22%
- Reduced model training time 70% via distributed PyTorch DDP on 8-GPU cluster (AWS p3.16xlarge)
- Mentored 4 junior data scientists; grew team ML maturity from ad-hoc scripts to reproducible ML pipelines

DataScale Systems | Data Scientist | 2019 – 2021

- Trained ensemble models (XGBoost, LightGBM, Random Forest) for real-time fraud detection — 99.2% AUC-ROC
- Engineered features from 1 TB+ daily Kafka streams using PySpark on AWS EMR
- Built A/B testing framework supporting 50+ concurrent experiments with correct CUPED variance reduction
- Applied causal inference (DiD, propensity-score matching) to measure true lift from product interventions
- NeurIPS 2020 publication: "Scalable Fraud Detection with Streaming Feature Engineering"

Carnegie Mellon Research Lab | ML Research Scientist | 2017 – 2019

- Researched multi-modal deep learning architectures; 4 publications at ICML (×2) and NeurIPS (×2)
- Developed open-source PyTorch extension library (600+ GitHub stars)
- Supervised 3 MSc students on applied ML research projects

EDUCATION

PhD in Machine Learning (Computer Science) | Carnegie Mellon University | 2019
Master of Science in Statistics | University of Chicago | 2015
Bachelor of Science in Mathematics | MIT | 2013

SKILLS

Python (expert), PyTorch, TensorFlow/Keras, scikit-learn, XGBoost, LightGBM, CatBoost, PySpark, SQL, R, Deep Learning (CNN, RNN, LSTM, Transformer, BERT, GPT fine-tuning), NLP, Computer Vision, Reinforcement Learning, Bayesian Methods, Statistical Modelling, Causal Inference, A/B Testing & Experimentation, Feature Engineering, MLflow, Kubeflow, Docker, Kubernetes, AWS SageMaker, GCP Vertex AI, Hadoop, Kafka, Git, Jupyter

PROJECTS & ACHIEVEMENTS

Kaggle: Gold medal in Tabular Playground Series (top 0.3%); Kaggle Master rank.
Research: 5 papers at NeurIPS / ICML with 300+ combined citations.
Open source: PyTorchExtensions — modular training utilities (600+ GitHub stars).
Production: Recommendation engine serving 50 M daily users with <50 ms p99 latency.

CERTIFICATIONS

AWS Certified Machine Learning – Specialty (2022), Google Professional Machine Learning Engineer (2023), Deep Learning Specialization – Coursera / Andrew Ng (2018)